

# The Language of Medicine

## 13<sup>th</sup> edition

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# Chapter 1

## Basic Word Structure

# Chapter Goals

- Identify basic objectives to guide your study of the medical language.
- Divide medical words into their component parts.
- Learn the meanings of basic combining forms, prefixes, and suffixes of the medical language.
- Use these combining forms, suffixes, and prefixes to build medical words.

# Chapter 1

## Lesson 1.1

# Objectives in Studying The Medical Language

- Analyze words by dividing them into component parts.
- Relate the medical terms to the structure and function of the human body.
- Be aware of spelling and pronunciation problems.

# QUICK QUIZ (Slide 1 of 5)

**1. All medical terms have:**

- A. a combining vowel
- B. two roots
- C. a suffix
- D. a prefix

# QUICK QUIZ (Slide 2 of 5)

**2. Which element appears at the beginning of a medical term?**

- A. Suffix
- B. Prefix
- C. Combining vowel
- D. Combining form

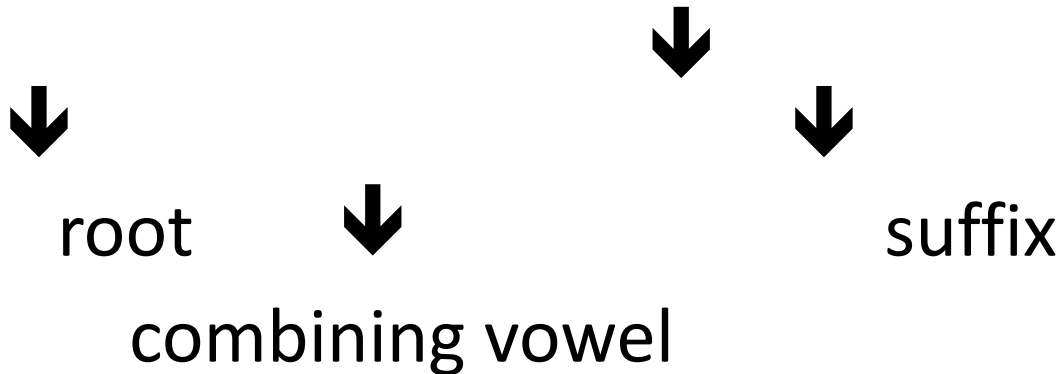
# Word Analysis (Slide 1 of 14)

**HEMAT/O/LOGY**



# Word Analysis (Slide 2 of 14)

**HEMAT/O/LOGY**



**hemat means blood**

**-logy means process of study**

# Word Analysis (Slide 3 of 14)

## Important elements:

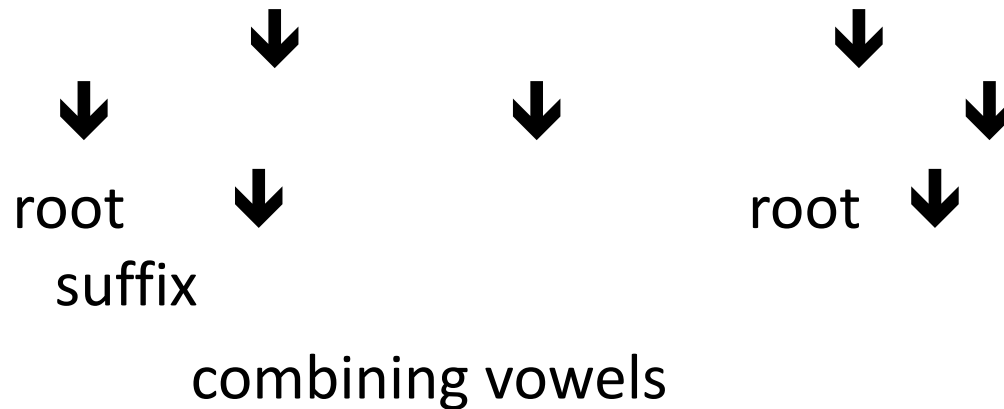
- **Root:** foundation of the word
- **Suffix:** word ending
- **Prefix:** small part attached to the beginning of a term
- **Combining vowel:** vowel (usually o) that links the root to the suffix or the root to another root
- **Combining form:** root + combining vowel

# Word Analysis (Slide 4 of 14)

**ELECTR/O/CARDI/O/GRAM**

# Word Analysis (Slide 5 of 14)

**ELECTR/O/CARDI/O/GRAM**



**electr means electricity**

**cardi means heart**

**-gram means record**

# Word Analysis (Slide 6 of 14)

## **GASTRITIS**

# Word Analysis (Slide 7 of 14)

GASTRITIS

GASTR/ITIS



root

suffix

**gastr-** means stomach

**-itis** means inflammation

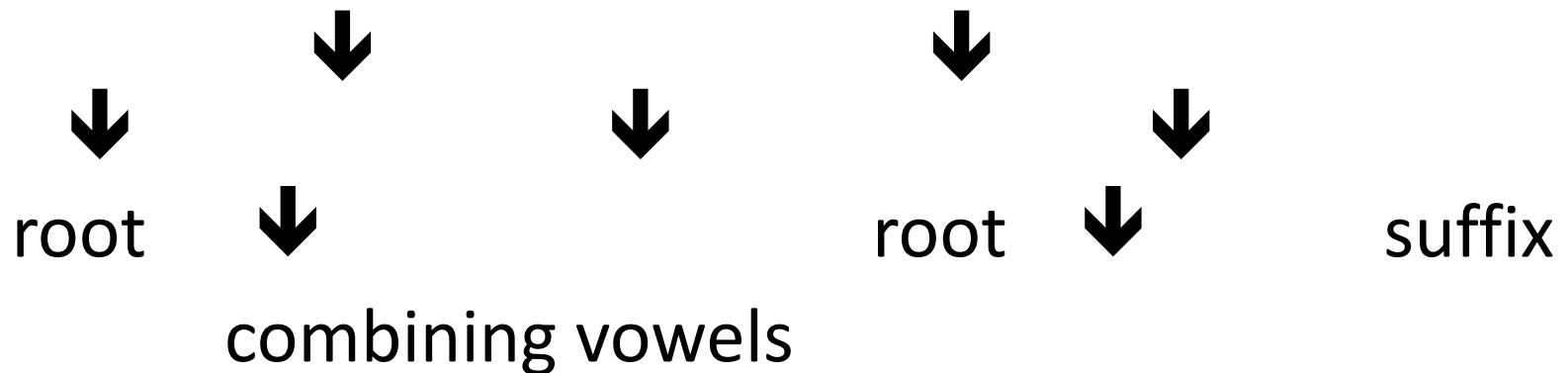
***Note:*** Drop combining vowel when suffix starts with a vowel  
(as in “itis”)

# Word Analysis (Slide 8 of 14)

**GASTR/O/ENTER/O/LOGY**

# Word Analysis (Slide 9 of 14)

**GASTR/O/ENTER/O/LOGY**



**gastr means stomach**

**enter means intestines**

**-logy means process of study**



### THREE General Rules:

1. READ the meaning of medical terms from the **suffix back to the beginning** of the term and across
2. DROP the combining vowel (usually o) before a suffix beginning with a vowel:  
**gastritis** *not* “**gastroitis**”
3. KEEP the combining vowel between two roots: **gastroenterology** *not* “**gastreenterology**”

# Word Analysis (Slide 11 of 14)

**HYPO/GASTR/IC**

# Word Analysis (Slide 12 of 14)

**HYPO/GASTR/IC**



prefix

root

suffix

**hypo- means below**

**gastr means stomach**

**-ic means pertaining to**

# Word Analysis (Slide 13 of 14)

**EPI/GASTR/IC**

# Word Analysis (Slide 14 of 14)

**EPI/GASTR/IC**



prefix   root   suffix

**epi-   means   above**

**gastr means   stomach**

**-ic   means   pertaining to**

# Combining Forms (Slide 1 of 16)

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## Combining Form

**aden/o**

**arthr/o**

**bi/o**

**carcin/o**

**cardi/o**

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# Combining Forms (Slide 2 of 16)

| Combining Form | Meaning           |
|----------------|-------------------|
| aden/o         | gland             |
| arthr/o        | joint             |
| bi/o           | life              |
| carcin/o       | cancerous, cancer |
| cardi/o        | heart             |

# QUICK QUIZ (Slide 3 of 5)

**3. Which term means inflammation of the gland?**

- A. Arthritis
- B. Adenitis
- C. Adenoma
- D. Cephalic



# Combining Forms (Slide 3 of 16)

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## Combining Form

**cephal/o**

**cerebr/o**

**cis/o**

**crin/o**

**cyst/o**

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# Combining Forms (Slide 4 of 16)

| Combining Form  | Meaning         |
|-----------------|-----------------|
| <b>cephal/o</b> | head            |
| <b>cerebr/o</b> | cerebrum        |
| <b>cis/o</b>    | to cut          |
| <b>crin/o</b>   | to secrete      |
| <b>cyst/o</b>   | urinary bladder |

# Combining Forms (Slide 5 of 16)

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## Combining Form

**cyt/o**

**derm/o or dermat/o**

**electr/o**

**encephal/o**

**enter/o**

# Combining Forms (Slide 6 of 16)

| Combining Form            | Meaning     |
|---------------------------|-------------|
| <b>cyt/o</b>              | cell        |
| <b>derm/o or dermat/o</b> | skin        |
| <b>electr/o</b>           | electricity |
| <b>encephal/o</b>         | brain       |
| <b>enter/o</b>            | intestines  |

# **QUICK QUIZ** (Slide 4 of 5)

**4. Which term means study of cells?**

- A. Cystoscopy
- B. Cystitis
- C. Cytology
- D. Hematology

# Combining Forms (Slide 7 of 16)

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## Combining Form

**erythr/o**

**gastr/o**

**glyc/o**

**gnos/o**

**gynec/o**

**hemat/o or hem/o**

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# Combining Forms (Slide 8 of 16)

| Combining Form          | Meaning       |
|-------------------------|---------------|
| <b>erythr/o</b>         | red           |
| <b>gastr/o</b>          | stomach       |
| <b>glyc/o</b>           | sugar         |
| <b>gnos/o</b>           | knowledge     |
| <b>gynec/o</b>          | woman, female |
| <b>hemat/o or hem/o</b> | blood         |

# Combining Forms (Slide 9 of 16)

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## Combining Form

hepat/o

iatr/o

leuk/o

log/o

nephr/o

neur/o

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# Combining Forms (Slide 10 of 16)

| Combining Form | Meaning              |
|----------------|----------------------|
| hepat/o        | liver                |
| iatr/o         | treatment, physician |
| leuk/o         | white                |
| log/o          | study of             |
| nephr/o        | kidney               |
| neur/o         | nerve                |

# Combining Forms (Slide 11 of 16)

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## Combining Form

**onc/o**

**ophthalm/o**

**oste/o**

**path/o**

**ped/o**

# Combining Forms (Slide 12 of 16)

| Combining Form    | Meaning |
|-------------------|---------|
| <b>onc/o</b>      | tumor   |
| <b>ophthalm/o</b> | eye     |
| <b>oste/o</b>     | bone    |
| <b>path/o</b>     | disease |
| <b>ped/o</b>      | child   |

# Combining Forms (Slide 13 of 16)

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## Combining Form

**psych/o**

**radi/o**

**ren/o**

**rhin/o**

**sarc/o**

# Combining Forms (Slide 14 of 16)

| Combining Form | Meaning |
|----------------|---------|
| psych/o        | mind    |
| radi/o         | x-rays  |
| ren/o          | kidney  |
| rhin/o         | nose    |
| sarc/o         | flesh   |

# Combining Forms (Slide 15 of 16)

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## Combining Form

**sect/o**

**thromb/o**

**ur/o**

# Combining Forms (Slide 16 of 16)

| Combining Form | Meaning              |
|----------------|----------------------|
| sect/o         | to cut               |
| thromb/o       | clot, clotting       |
| ur/o           | urinary tract, urine |

# Suffixes (Slide 1 of 10)

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## Suffix

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**-ac**

**-al**

**-algia**

**-cyte**

**-ectomy**

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# Suffixes (Slide 2 of 10)

| Suffix         | Meaning           |
|----------------|-------------------|
| <b>-ac</b>     | pertaining to     |
| <b>-al</b>     | pertaining to     |
| <b>-algia</b>  | pain              |
| <b>-cyte</b>   | cell              |
| <b>-ectomy</b> | excision, removal |

## QUICK QUIZ (Slide 5 of 5)

**5. Which suffix means to “cut out” or remove?**

A. -ectomy

B. -scopy

C. -itis

D. -opsy

# Suffixes (Slide 3 of 10)

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## Suffix

**-emia**

**-genic**

**-globin**

**-gram**

**-ic, ical**

# Suffixes (Slide 4 of 10)

| Suffix    | Meaning           |
|-----------|-------------------|
| -emia     | blood condition   |
| -genic    | produced by or in |
| -globin   | protein           |
| -gram     | record            |
| -ic, ical | pertaining to     |

# Suffixes (Slide 5 of 10)

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## Suffix

**-ion**

**-ist**

**-itis**

**-logy**

**-oma**

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# Suffixes (Slide 6 of 10)

| Suffix       | Meaning               |
|--------------|-----------------------|
| <b>-ion</b>  | process               |
| <b>-ist</b>  | specialist            |
| <b>-itis</b> | inflammation          |
| <b>-logy</b> | process of study      |
| <b>-oma</b>  | tumor, mass, swelling |

# Suffixes (Slide 7 of 10)

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## Suffix

**-opsy**

**-osis**

**-pathy**

**-scope**

**-scopy**

# Suffixes (Slide 8 of 10)

| Suffix        | Meaning                        |
|---------------|--------------------------------|
| <b>-opsy</b>  | process of viewing             |
| <b>-osis</b>  | condition, usually abnormal    |
| <b>-pathy</b> | disease condition              |
| <b>-scope</b> | instrument to visually examine |
| <b>-scopy</b> | process of visually examining  |



# Suffixes (Slide 9 of 10)

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## **Suffix**

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**-sis**

**-tomy**

**-y**

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# Suffixes (Slide 10 of 10)

| Suffix       | Meaning                      |
|--------------|------------------------------|
| <b>-sis</b>  | state of; condition          |
| <b>-tomy</b> | process of cutting, incision |
| <b>-y</b>    | process, condition           |

# Prefixes (Slide 1 of 6)

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## Prefix

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**a-, an-**

**aut-, auto-**

**dia-**

**end-, endo-**

**epi-**

**ex-, exo-**

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# Prefixes (Slide 2 of 6)

| Prefix             | Meaning                  |
|--------------------|--------------------------|
| <b>a-, an-</b>     | no, not, without         |
| <b>aut-, auto</b>  | self, own                |
| <b>dia-</b>        | complete, through        |
| <b>end-, endo-</b> | within                   |
| <b>epi-</b>        | above, upon              |
| <b>ex-, exo-</b>   | out, outside of, outward |

# Prefixes (Slide 3 of 6)

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## Prefix

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**hyper-**

**hypo-**

**in-**

**peri-**

**pro-**

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# Prefixes (Slide 4 of 6)

| Prefix        | Meaning                                   |
|---------------|-------------------------------------------|
| <b>hyper-</b> | excessive, above, more than normal        |
| <b>hypo-</b>  | deficient, below, under, less than normal |
| <b>in-</b>    | into, in                                  |
| <b>peri-</b>  | surrounding, around                       |
| <b>pro-</b>   | before, forward                           |

# Prefixes (Slide 5 of 6)

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**Prefix**

**re-**

**retro-**

**sub-**

**trans-**

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# Prefixes (Slide 6 of 6)

| Prefix | Meaning               |
|--------|-----------------------|
| re-    | back, backward, again |
| retro- | behind                |
| sub-   | below, under          |
| trans- | across, through       |